

Supplementary Material: Causal Intervention for Hallucination-Resistant Medical Visual Question Answering

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Appendix A

Formalization of Assumption 1 and Logit Contrast Cancellation

We formalize the medical diagnostic reasoning process under the Structural Causal Model (SCM) over five primary causal random variables: the input medical scan $I \in \mathcal{I}$, the natural language clinical query $Q \in \mathcal{Q}$, the unobserved dataset-level confounder $C \in \mathcal{C}$ (capturing institutional reporting templates, scanner protocol co-occurrences, and demographic priors), the extracted visual patch representations $\mathbf{V} = f_\theta(I) \in \mathcal{V}$, and the predicted diagnostic answer $A \in \mathcal{A}$, along with the candidate lesion mask $\mathbf{M} \in [0, 1]^{H \times W}$ and intervened counterfactual scan I_{cf} .

In the visual representation space, the visual tokens $\mathbf{V}$ decompose into candidate focal lesion representations $\mathbf{V}_{focal} = \mathbf{V} \odot \mathbf{M}_{patch}$ and surrounding anatomical background representations $\mathbf{V}_{bg} = \mathbf{V} \odot (1 - \mathbf{M}_{patch})$. In conventional observational Med-VQA, models parameterize a joint logit function $\mathbf{L}(I, Q) \in \mathbb{R}^K$. Because observational training maximizes conditional likelihood $P(A \mid I, Q)$ without graph intervention, the unblocked backdoor path $I \leftarrow C \rightarrow A$ and direct language correlations induce additive non-causal shortcuts:

$$\mathbf{L}(I, Q) = \mathbf{g}(\mathbf{V}_{focal}, Q) + \mathbf{h}_{anat}(\mathbf{V}_{bg}, Q) + \mathbf{s}(C, Q) + \epsilon, \quad (1)$$

where $\mathbf{g}(\cdot)$ represents the authentic causal effect of localized focal pathology, $\mathbf{h}_{anat}(\cdot)$ represents the authentic diagnostic evidence contributed by anatomical organ substrates, $\mathbf{s}(C, Q)$ encapsulates the spurious confounding induced by scanner characteristics and language prior shortcuts, and ϵ is zero-mean residual noise.

Under Assumption 1 (Localized Pathological Causality, formalized in main text Section III-A), authentic visual evidence for a focal lesion is strictly confined to $\mathbf{M} \odot I$, whereas dataset-level confounding operates through background image characteristics $I_{bg} = (1 - \mathbf{M}) \odot I$ and language reporting templates Q . When applying the physical counterfactual intervention $do(I = I_{cf})$, the generative inpainter synthesizes healthy tissue within the localized candidate region $\mathbf{M}$ while leaving background anatomy strictly unaltered:

$$I_{cf} = (1 - \mathbf{M}) \odot I + \mathbf{M} \odot G_\phi(I, \mathbf{M}). \quad (2)$$

Because the candidate lesion is replaced with synthesized healthy tissue, focal pathological evidence is neutral-

ized ($\mathbf{g}(\mathbf{V}_{cf, focal}, Q) \approx \mathbf{0}$). Meanwhile, the background anatomy $(1 - \mathbf{M}) \odot I$, the visual background tokens $\mathbf{V}_{bg}$, and the confounder C and query Q remain identical between I and I_{cf} . Evaluating the model on the counterfactual scan yields:

$$\mathbf{L}(I_{cf}, Q) = \mathbf{h}_{anat}(\mathbf{V}_{bg}, Q) + \mathbf{s}(C, Q) + \epsilon'. \quad (3)$$

Subtracting the counterfactual pass with dynamic scaling $\gamma(Q) \approx 1$ isolates the focal causal contrast:

$$\begin{aligned} \mathbf{L}_{focal} &= \mathbf{L}(I, Q) - \gamma(Q)\mathbf{L}(I_{cf}, Q) \\ &= \mathbf{g}(\mathbf{V}_{focal}, Q) + (1 - \gamma(Q)) [\mathbf{h}_{anat}(\mathbf{V}_{bg}, Q) + \mathbf{s}(C, Q)] \\ &\quad + (\epsilon - \gamma(Q)\epsilon'). \end{aligned} \quad (4)$$

When $\gamma(Q) \approx 1.0$ (or in the canonical baseline where $\gamma(Q) = 1.0$), the shared confounding term $\mathbf{s}(C, Q)$ and the shared background signal cancel analytically ($(1 - \gamma(Q)) [\mathbf{h}_{anat} + \mathbf{s}] \approx \mathbf{0}$), cleanly isolating the authentic focal pathological effect $\mathbf{g}(\mathbf{V}_{focal}, Q)$. In our neural implementation, $\gamma(Q) \in \mathbb{R}^+$ is parameterized via Softplus and initialized with projection bias b_γ such that $\gamma(Q) \approx 1.0$. Importantly, $\gamma(Q)$ operates over strictly positive real values $\mathbb{R}^+$ rather than being constrained to the unit interval $[0, 1]$, providing the network unconstrained headroom to scale contrast adaptively across varying prompt complexities while ensuring strictly positive subtraction ($\gamma(Q) > 0$).

Conversely, for anatomical queries where the question evaluates broader organ substrates (such that $\mathbf{g}(\mathbf{V}_{focal}, Q) = \mathbf{0}$), both passes contain identical anatomical signal ($\mathbf{L}(I, Q) \approx \mathbf{L}(I_{cf}, Q)$). Setting $\gamma(Q) \approx 1.0$ under uniform naive subtraction would force $(1 - \gamma(Q))\mathbf{h}_{anat} \approx \mathbf{0}$, triggering the contrast cancellation dilemma by destroying the valid anatomical evidence needed for the answer. Under the proposed APD-CC architecture, the continuous Semantic Causal Router $\alpha(Q) \in [0, 1]$ (parameterized via a Sigmoid activation, representing a continuous interpolation weight over $[0, 1]$ rather than a hard discrete $\{0, 1\}$ decision) outputs $\alpha(Q) \approx 0$ on anatomical queries. This smoothly routes such queries directly to Anatomical Substrate Evidence Anchoring:

$$\mathbf{L}_{anat} = \mathbf{L}(I, Q) + \beta(Q) \cdot \mathbf{E}_{anat}, \quad (5)$$

where $\mathbf{E}_{anat} = \mathbf{W}_e \cdot \text{MeanPool}(\mathbf{V} \odot (1 - \mathbf{M}_{patch}))$ directly reinforces verified visual anatomical tokens $\mathbf{V}_{bg}$ without destructive contrast subtraction (with dynamic factor

$\beta(Q) \in \mathbb{R}^+$ also parameterized via Softplus). The convex combination $\mathbf{L}^* = \alpha(Q)\mathbf{L}_{\text{focal}} + (1 - \alpha(Q))\mathbf{L}_{\text{anat}}$ thus adaptively balances focal debiasing against anatomical evidence anchoring across query types without signal collapse.

Appendix B

Hyperparameter Specifications and Training Regimes

All neural backbones and causal heads were trained using PyTorch 2.x across NVIDIA GPUs (T4, P100, and A100). Multi-task loss weights ($\lambda_{\text{calib}}, \lambda_{\text{con}}, \lambda_{\text{percep}}, \lambda_{\text{style}}$) and optimization parameters were tuned via grid search evaluated strictly on held-out validation folds. Table S1 outlines the complete training hyperparameter configuration.

TABLE S1
Comprehensive Hyperparameter Configuration

Hyperparameter	Operational Setting
Visual Encoder	Vision Transformer (vit-base-patch16-224-in21k)
Text Encoder	PubMedBERT (BiomedBERT-base-uncased)
Counterfactual Inpainter	Micro U-Net (Convolutional Context-Encoder)
Image Resolution	$224 \times 224 \times 3$ pixels
Batch Size	16
Backbone Learning Rate	2.5×10^{-5} (fine-tuned)
Fusion & Head Learning Rate	5.0×10^{-4} (and for G_ϕ in Stage 1; trained from scratch)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Weight Decay	1.0×10^{-2}
LR Scheduler	Cosine Annealing with 2-epoch linear warmup
Multi-Task Loss Weights (Stage 2)	$\lambda_{\text{calib}} = 0.2, \lambda_{\text{con}} = 0.1$
Inpainting Loss Weights (Stage 1)	$\mathcal{L}_{\text{L1}} = 1.0, \lambda_{\text{percep}} = 0.1, \lambda_{\text{style}} = 250.0$
Causal Dynamic Scale Init	$\gamma(Q) \approx 1.0$ (via initialized projection bias b_γ)
Decision Gate Threshold τ_1	0.65 (Screening operational point)
Decision Gate Threshold τ_2	0.85 (Triage referral operational point)
Evaluation Random Seeds	{42, 43, 44} (fixed multi-seed splits)

A. Generative Counterfactual Inpainter (G_ϕ) Architecture and Stage 1 Pre-Training Protocol

To ensure complete experimental reproducibility of the physical generative intervention $do(I = I_{\text{cf}})$, we provide the detailed architecture and training protocols for the Generative Counterfactual Inpainter G_ϕ :

1) Network Topology: G_ϕ is structured as a lightweight convolutional Micro U-Net ($\approx 1.25 \times 10^6$ parameters) taking a 4-channel concatenated input $[I; \mathbf{M}] \in \mathbb{R}^{4 \times 224 \times 224}$ (the 3-channel medical scan and the 1-channel spatial mask). The encoder consists of four convolutional downsampling blocks with feature channel depths {32, 64, 128, 256}, using 3×3 kernels, stride 2, Instance Normalization, and LeakyReLU activations ($\alpha = 0.2$). The bottleneck comprises two residual blocks with dilated convolutions ($d = 2, 256$ channels) to expand the receptive field without reducing spatial resolution. The decoder mirrors the encoder with four transposed convolutional upsampling blocks ({128, 64, 32, 16} channels, stride 2) and concatenation skip connections from the corresponding encoder stages. A final 1×1 convolution with Tanh

activation outputs the synthesized image. This compact design maintains an average inference latency of only 22.3ms on a single GPU.

2) Stage 1 Pre-Training Cohorts: Self-supervised pre-training is conducted strictly on non-pathological, normal scans sampled exclusively from the official training splits of the benchmark cohorts (e.g., normal radiographs in VQA-RAD and SLAKE, and benign histological tissue in PathVQA). Crucially, validation and test splits are strictly held out to eliminate any risk of data leakage.

3) Synthetic Mask Generation ($\mathbf{M}_{\text{rand}}$): During Stage 1, random synthetic masks $\mathbf{M}_{\text{rand}} \in \{0, 1\}^{H \times W}$ are generated dynamically on-the-fly to simulate plausible lesion topologies: (i) random rectangular bounding boxes covering 10% to 25% of the scan area, and (ii) irregular free-form strokes (1–4 continuous brush strokes, line width 8–25 pixels).

4) Pre-Training Objective and Optimization: Pre-training optimizes the composite reconstruction objective:

$$\mathcal{L}_{\text{inp}} = \mathcal{L}_{\text{L1}} + \lambda_{\text{percep}} \mathcal{L}_{\text{percep}} + \lambda_{\text{style}} \mathcal{L}_{\text{style}}, \quad (6)$$

where $\mathcal{L}_{\text{L1}} = \|G_\phi(I, \mathbf{M}_{\text{rand}}) - I\|_1$ guarantees pixel-level fidelity and boundary continuity; perceptual loss $\mathcal{L}_{\text{percep}} = \sum_{l=1}^3 \frac{1}{C_l H_l W_l} \|\psi_l(G_\phi(I, \mathbf{M}_{\text{rand}})) - \psi_l(I)\|_1$ evaluates multi-scale activations from a fixed ImageNet-pretrained VGG-16 network (layers relu1_2, relu2_2, relu3_3); and style loss $\mathcal{L}_{\text{style}} = \sum_{l=1}^3 \|\mathcal{G}(\psi_l(G_\phi(I, \mathbf{M}_{\text{rand}}))) - \mathcal{G}(\psi_l(I))\|_F^2$ penalizes Gram-matrix correlation discrepancies. Training is executed using AdamW (learning rate 5.0×10^{-4} , weight decay 0.01) for 50 epochs with cosine decay and a batch size of 16. During Stage 2 end-to-end multimodal training, G_ϕ remains completely frozen.

Appendix C

Selective Abstention and Validation Threshold Calibration

To prevent optimistic evaluation bias, selective classification thresholds $\tau \in [0, 1]$ are calibrated and frozen strictly on validation folds prior to test set evaluation [1], [2]. Following clinical risk-coverage principles, on each validation fold the threshold τ is chosen to satisfy target selective risk constraints ($\text{Risk}(\tau) \leq \epsilon$) across two operational clinical deployment regimes:

- 1) High-Throughput Screening Regime ($\tau_1 = 0.65$): Calibrated on validation splits to maximize patient coverage while bounding selective diagnostic error below $\approx 10\%$, admitting broad automated throughput.
- 2) Conservative Clinical Triage Regime ($\tau_2 = 0.85$): Calibrated on validation splits to enforce stringent safety bounds on accepted decisions, targeting a clinical error rate $\leq 2.5\%$ by safely referring ambiguous, motion-corrupted, or out-of-distribution queries to clinical specialists.

Empirical Selective Risk $\text{Risk}(\tau)$, Patient Coverage $\text{Coverage}(\tau)$, and Specialist Referral Rate $\text{Referral}(\tau) =$

$1 - \text{Coverage}(\tau)$ are evaluated on test sets under the frozen validation thresholds:

$$\text{Risk}(\tau) = \frac{\sum_{i=1}^N \mathbf{1}(\hat{A}_i \neq Y_i^*) \cdot \mathbf{1}(P_{\max,i} \geq \tau)}{\sum_{i=1}^N \mathbf{1}(P_{\max,i} \geq \tau)}, \quad (7)$$

$$\text{Coverage}(\tau) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(P_{\max,i} \geq \tau), \quad (8)$$

where $P_{\max,i} = \max_a P_{\text{causal}}(a \mid I_i, Q_i)$ and $\hat{A}_i = \arg \max_a P_{\text{causal}}(a \mid I_i, Q_i)$ denotes the model’s top predicted answer. Table S3 summarizes the complete operational schedule.

Appendix D

Extended Uncertainty Calibration and Proper Scoring Evaluation

Table S2 provides the comprehensive multi-cohort calibration and proper scoring breakdown, reporting Maximum Calibration Error (MCE), Negative Log-Likelihood (NLL), and Weighted-Averaged F1 alongside primary Expected Calibration Error (ECE), Brier Score, and Macro-F1 across the four Med-VQA benchmark domains (VQA-RAD, SLAKE, PathVQA, and Kvasir-VQA-x1).

Appendix E

Selective Abstention Operating Points and Clinical Risk-Coverage

Table S3 summarizes the full selective classification schedule evaluating empirical selective risk ($\text{Risk}(\tau)$), patient coverage ($\text{Coverage}(\tau)$), and algorithmic deferral rates ($\text{Referral}(\tau) = 1 - \text{Coverage}(\tau)$) across high-throughput screening ($\tau_1 = 0.65$) and conservative diagnostic triage ($\tau_2 = 0.85$) operational thresholds in our retrospective benchmark evaluation.

Appendix F

Fine-Grained POPE Visual Hallucination Probing Breakdown

Table S4 details the comprehensive Polling-based Object Probing Evaluation (POPE) results across Random, Popular, and Adversarial probe sets, reporting binary classification accuracy, precision, recall, F1-score, affirmative response ratio (Yes-Ratio), and empirical hallucination rate.

Appendix G

Mathematical Formulations of Secondary Metrics and Statistical Testing Protocols

This section provides complete mathematical formulations for all secondary evaluation metrics and the formal statistical hypothesis testing protocols employed across the manuscript.

A. Secondary Probabilistic Calibration and Scoring Metrics

1) Maximum Calibration Error (MCE): While ECE measures the average divergence across probability bins, MCE quantifies the worst-case deviation between empirical accuracy and predicted confidence across all $M = 10$ quantile bins:

$$\text{MCE} = \max_{m \in \{1, \dots, M\}} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (9)$$

In safety-critical clinical applications, a bounded MCE guarantees that the model does not exhibit severe localized miscalibration pockets even in low-density confidence regimes.

2) Negative Log-Likelihood (NLL): Serving as a standard proper scoring rule, NLL evaluates the average log loss over the true class probabilities across N evaluation samples and K vocabulary classes:

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log(P_{\text{causal}}(a_k \mid I_i, Q_i)). \quad (10)$$

3) Weighted-Averaged F1-Score ($\text{F1}_{\text{weighted}}$): Complementing the unweighted Macro-F1 (which assigns equal weight to each class), Weighted-F1 weights each class by its empirical prevalence $N_k = \sum_{i=1}^N y_{i,k}$:

$$\text{F1}_{\text{weighted}} = \sum_{k=1}^K \frac{N_k}{N} \text{F1}_k, \quad \text{F1}_k = \frac{2 \cdot \text{Precision}_k \cdot \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \quad (11)$$

B. Spatial Grounding and Attribution Metrics

1) Pointing Game Accuracy (PG): For an extracted continuous ROI heatmap $\mathbf{M} \in [0, 1]^{H \times W}$ and ground-truth radiologist bounding box $\mathcal{B}_{\text{gt}}$, a hit is scored if the maximum activation coordinate falls within the annotation:

$$\text{PG} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(\arg \max_{(u,v)} \mathbf{M}_i(u, v) \in \mathcal{B}_{\text{gt},i} \right). \quad (12)$$

2) Mean Intersection-over-Union (mIoU) and Dice Score (DSC): Binarizing $\mathbf{M}$ at threshold $\theta = 0.5$ yields binary prediction $\mathbf{M}_{\text{bin}} \in \{0, 1\}^{H \times W}$:

$$\begin{aligned} \text{mIoU} &= \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{M}_{\text{bin},i} \cap \mathcal{B}_{\text{gt},i}|}{|\mathbf{M}_{\text{bin},i} \cup \mathcal{B}_{\text{gt},i}|}, \\ \text{DSC} &= \frac{1}{N} \sum_{i=1}^N \frac{2|\mathbf{M}_{\text{bin},i} \cap \mathcal{B}_{\text{gt},i}|}{|\mathbf{M}_{\text{bin},i}| + |\mathcal{B}_{\text{gt},i}|}. \end{aligned} \quad (13)$$

3) Spatial Attribution Consistency (Att. Cons.): Quantifies attribution invariance under multi-crop spatial transformations $\mathcal{T}$. For an input image I and perturbed view $\mathcal{T}(I)$, attribution consistency evaluates the cosine similarity of spatial heatmaps aligned via inverse coordinate mapping $\mathcal{T}^{-1}$:

$$\text{Att.-Cons.} = \frac{\langle \mathbf{M}(I), \mathcal{T}^{-1}(\mathbf{M}(\mathcal{T}(I))) \rangle}{\|\mathbf{M}(I)\|_2 \cdot \|\mathcal{T}^{-1}(\mathbf{M}(\mathcal{T}(I)))\|_2}. \quad (14)$$

TABLE S2
Extended Diagnostic Calibration and Probabilistic Scoring Across Multi-Modal Clinical Benchmarks

Dataset	Model Architecture	Top-1 Acc (%) $\uparrow$	ECE $\downarrow$	MCE $\downarrow$	Brier Score $\downarrow$	NLL $\downarrow$	Macro F1 $\uparrow$	Weighted F1 $\uparrow$
VQA-RAD	Baseline VLM (ViT + PubMedBERT)	92.58% $\pm$ 0.42%	0.0195 $\pm$ 0.0035	0.0842 $\pm$ 0.0081	0.0887 $\pm$ 0.0028	0.2450 $\pm$ 0.0062	0.856 $\pm$ 0.006	0.918 $\pm$ 0.004
	Proposed CI-GCI	92.41% $\pm$ 0.48%	0.0158 $\pm$ 0.0024	0.0612 $\pm$ 0.0054	0.0894 $\pm$ 0.0025	0.2312 $\pm$ 0.0048	0.855 $\pm$ 0.007	0.920 $\pm$ 0.005
SLAKE	Baseline VLM (ViT + PubMedBERT)	74.80% $\pm$ 0.32%	0.1852 $\pm$ 0.0041	0.3240 $\pm$ 0.0112	0.3640 $\pm$ 0.0062	0.7420 $\pm$ 0.0125	0.684 $\pm$ 0.005	0.732 $\pm$ 0.004
	Proposed CI-GCI (APD-CC)	90.60% $\pm$ 0.38%	0.0195 $\pm$ 0.0012	0.0785 $\pm$ 0.0048	0.1084 $\pm$ 0.0016	0.2840 $\pm$ 0.0055	0.882 $\pm$ 0.004	0.902 $\pm$ 0.003
PathVQA	Baseline VLM (ViT + PubMedBERT)	89.77% $\pm$ 0.13%	0.0764 $\pm$ 0.0004	0.1980 $\pm$ 0.0065	0.1868 $\pm$ 0.0008	0.4120 $\pm$ 0.0078	0.897 $\pm$ 0.001	0.898 $\pm$ 0.001
	Proposed CI-GCI	89.48% $\pm$ 0.27%	0.0792 $\pm$ 0.0022	0.1740 $\pm$ 0.0052	0.1961 $\pm$ 0.0012	0.4050 $\pm$ 0.0065	0.894 $\pm$ 0.003	0.895 $\pm$ 0.002
Kvasir-VQA-x1	Baseline VLM (ViT + PubMedBERT)	76.20% $\pm$ 0.45%	0.1420 $\pm$ 0.0035	0.2850 $\pm$ 0.0094	0.3180 $\pm$ 0.0050	0.6850 $\pm$ 0.0110	0.710 $\pm$ 0.004	0.755 $\pm$ 0.004
	Proposed CI-GCI	82.40% $\pm$ 0.38%	0.0234 $\pm$ 0.0012	0.0910 $\pm$ 0.0045	0.1380 $\pm$ 0.0018	0.3420 $\pm$ 0.0062	0.806 $\pm$ 0.003	0.819 $\pm$ 0.003
Benchmark Average	Baseline VLM (ViT + PubMedBERT)	83.34%	0.1058	0.2228	0.2394	0.5210	0.787	0.826
	Proposed CI-GCI	88.72%	0.0345	0.1012	0.1330	0.3156	0.859	0.884

All metrics reported as mean $\pm$ standard deviation across 3 random seeds (42, 43, 44). Differences on SLAKE and Kvasir-VQA-x1 are statistically significant ($p < 0.001$, McNemar's test). Diagnostic parity on VQA-RAD and PathVQA closed splits confirmed via Two One-Sided Tests (TOST, $\delta = 0.02, p_{\text{TOST}} < 0.05$). Bold indicates best or equivalent performance.

TABLE S3
Decision-Theoretic Selective Triage Operating Schedule

Operational Regime	Threshold (τ)	Coverage (%)	Selective Risk (%) $\downarrow$	Referral Rate (%)
Baseline VLM (Uncalibrated)	$\tau = 0.00$	100.0%	25.20%	0.00%
Baseline VLM (Screening)	$\tau_1 = 0.65$	91.40%	18.40%	8.60%
Baseline VLM (Strict Triage)	$\tau_2 = 0.85$	74.20%	12.80%	25.80%
Proposed CI-GCI (Screening)	$\tau_1 = 0.65$	94.60%	6.10%	5.40%
Proposed CI-GCI (Clinical Triage)	$\tau_2 = 0.85$	72.50%*	2.40%	27.50%

Bold indicates superior performance at matched operating thresholds (higher automated patient coverage and lower selective diagnostic risk $\mathcal{R}(\tau)$). Referral rate represents the fraction of ambiguous cases referred for specialist review (Referral = 100% – Coverage). *Reported on the SLAKE multi-organ benchmark cohort; on less complex screening subsets, automated coverage reaches up to 85.12% at 2.45% risk (14.88% referral).

TABLE S4
Fine-Grained POPE Visual Hallucination Probing Evaluation

Probe Category	Model Architecture	Accuracy (%) $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1-Score $\uparrow$	Yes-Ratio	Hallucination Rate $\downarrow$
Random Probes	Baseline VLM (ViT + PubMedBERT)	79.8%	0.763	0.865	0.811	56.6%	0.268
	Proposed CI-GCI	94.8%	0.943	0.954	0.948	50.6%	0.058
Popular Probes	Baseline VLM (ViT + PubMedBERT)	73.7%	0.703	0.820	0.757	58.3%	0.346
	Proposed CI-GCI	91.6%	0.905	0.931	0.918	51.5%	0.098
Adversarial Probes	Baseline VLM (ViT + PubMedBERT)	69.7%	0.661	0.805	0.726	60.9%	0.412
	Proposed CI-GCI	88.5%	0.868	0.908	0.888	52.3%	0.138
Macro-Average / Overall	Baseline VLM (ViT + PubMedBERT)	74.4%	0.709	0.830	0.765	58.6%	0.342
	Proposed CI-GCI	91.6%	0.905	0.931	0.918	51.5%	0.098

Bold indicates superior performance across probe categories. The Macro-Average matches the canonical evaluation reported in main paper Table VI (0.342 $\rightarrow$ 0.098, representing an over 71% relative reduction in visual hallucinations).

C. Counterfactual Inpainting Fidelity Metrics

1) Background Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM): Evaluated strictly outside the candidate lesion mask ($1 - \mathbf{M}$) to verify near-lossless context retention:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}_{\text{bg}}} \right), \quad (15)$$

where $\text{MSE}_{\text{bg}} = \frac{1}{\sum (1 - \mathbf{M})} \sum_{u,v} (1 - \mathbf{M}(u,v)) \cdot (I(u,v) - I_{\text{cf}}(u,v))^2$. Structural similarity SSIM evaluates local luminance μ , contrast σ , and structural correlation across sliding windows over unmasked tissue:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}. \quad (16)$$

2) Distributional Realism via Patch ROC-AUC: To objectively test whether synthesized healthy patches introduce detectable synthesis seams, a linear patch discriminator is trained on a 50/50 mix of authentic healthy patches and synthetic inpainting patches. The discriminator's test Area

Under the ROC Curve (ROC-AUC) measures separability.

In this adversarial validation setting, an AUC near the chance baseline (≈ 0.500) indicates that synthetic patches are distributionally indistinguishable from natural tissue.

D. Formal Statistical Hypothesis Testing Protocols

1) Two One-Sided Tests (TOST) for Diagnostic Equivalence: To verify that counterfactual debiasing preserves strong observational baseline reasoning on closed queries without degradation, we employ Schuirmann's TOST procedure under an equivalence margin $\delta = 0.02$ and significance level $\alpha = 0.05$. Let $\hat{p}_1$ and $\hat{p}_2$ denote empirical accuracies of proposed CI-GCI and the baseline VLM on n paired test queries. Under McNemar paired contingency, let n_{10} count instances where baseline is correct and proposed is incorrect, and n_{01} count instances where proposed is correct and baseline is incorrect. The estimated difference is $\hat{d} = \hat{p}_1 - \hat{p}_2 = (n_{01} - n_{10})/n$, with

standard error:

$$\text{SE}(\hat{d}) = \frac{\sqrt{(n_{10} + n_{01}) - \frac{(n_{10} - n_{01})^2}{n}}}{n}. \quad (17)$$

The two one-sided null hypotheses are $H_{01} : d \leq -\delta$ (proposed is unacceptably inferior) and $H_{02} : d \geq +\delta$ (proposed is unacceptably divergent). The respective test statistics are:

$$Z_1 = \frac{\hat{d} - (-\delta)}{\text{SE}(\hat{d})}, \quad Z_2 = \frac{\delta - \hat{d}}{\text{SE}(\hat{d})}. \quad (18)$$

Equivalence is declared at significance $\alpha = 0.05$ if and only if $p_{\text{TOST}} = \max(1 - \Phi(Z_1), 1 - \Phi(Z_2)) < 0.05$, which is mathematically equivalent to the 90% confidence interval $[\hat{d} - z_{0.05}\text{SE}(\hat{d}), \hat{d} + z_{0.05}\text{SE}(\hat{d})]$ falling strictly inside $[-\delta, +\delta]$. On VQA-RAD closed queries ($n = 284$), for a representative evaluation seed, observed paired discordances are $n_{10} = 4, n_{01} = 4$ (both models achieving 263/284 correct), yielding $\hat{d} = 0.000$, $\text{SE} = 0.00996$, 90% CI $[-0.0164, +0.0164] \subset [-0.02, +0.02]$, and $Z_1 = Z_2 = 2.008$, confirming statistical equivalence ($p_{\text{TOST}} = 0.022 < 0.05$). Across three independent seeds, CI-GCI achieves a mean accuracy of $92.41\% \pm 0.48\%$ compared to the baseline’s $92.58\% \pm 0.42\%$ (mean paired difference -0.17%). The low discordance rate ($8/284 = 2.8\%$) is directly explained by the 202-of-203 train-test image leakage documented by Yang et al. [3], which induces both models to memorize identical closed-set shortcuts.

2) Non-Parametric Paired Bootstrap Intervals: For all reported benchmark statistics, metric uncertainty intervals are generated using $B = 2,000$ paired bootstrap resamples drawn with replacement. The empirical 2.5th and 97.5th percentiles define the two-sided 95% confidence interval.

3) McNemar’s Paired Significance Test with Continuity Correction: For paired binary classification comparisons between baseline and proposed models, paired discordances are evaluated using McNemar’s test with Edwards’ continuity correction [4]:

$$\chi^2 = \frac{(|n_{10} - n_{01}| - 1)^2}{n_{10} + n_{01}}, \quad (19)$$

with exact binomial test probabilities computed when discordant pairs $n_{10} + n_{01} < 25$.

4) Holm-Bonferroni Family-Wise Error Correction: When evaluating multiple hypothesis tests across the five benchmark domains simultaneously, p-values $\{p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}\}$ are ordered, and each test k is rejected if $p_{(k)} < \frac{\alpha}{m-k+1}$, controlling the family-wise error rate at $\alpha = 0.05$.

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